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	DOCUMENTED INFORMATION		Revision No.	00
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RESEARCH PROPOSAL				

Research Program/Project/Study Title: Development of a Dual Oyster Shucking Clamp

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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:	Roxas						
		Province:	Capiz						
		Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☐ Basic	☐ Pilot Testing								
☒ Applied	☐ Tech. Promotion/Commercialization								
(7) Sector/Commodity Fisheries/fish		(8) Discipline/Field of Specialization							

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Oysters are considered as one of the most valuable bivalve mollusks in the Philippines which produced over 53 thousand metric tons (MT) in 2020. These commodity is also considered to be a great source of protein and minerals vital to human nutrition and frequently served in restaurants and hotels (DOST-PCCARRD, 2020). In Capiz, oyster is also one of the best and most popular seafood dishes. Oysters are usually best enjoyed blanched or partially cooked. Some people eat oysters raw, but it's recommended to prepare them before consuming because they may still contain live organisms (Seafood Blogspot, 2017). Oysters contains a soft meat enclosed by a hard shell made of calcium carbonate, essentially rock, and laid down by the mantle of the oyster meat. It consists of two valves connected by a hinge ligament, and an adductor muscle that holds the oyster closed (Wheaton and Hall, 2007).

Shucking is the process of removing the meat from the oyster without causing any damage to the shell and meat. Traditionally, shucking has been performed manually by knowledgeable individuals known as "shuckers." The process of shucking is a difficult task that involves inserting a blade of some kind into the shell and cutting the adductor muscle attachments. The work is physically demanding and tedious, and there's a chance of danger if the shucking knife slips and pierces or slashes the shucker's hand (Wheaton and Hall, 2007). Under such circumstances, slicing open oysters with blades may result in cuts. Several tools have been created to assist the user in preventing injuries when shucking oysters and other bivalves (Nicholas, 2023). Furthermore, the fact that there is a severe lack of experienced shuckers, working condition is not nice, and a usual shucker has work only when there are oysters available are the main issues with manual shucking (Wheaton and Hall, 2007). Thus, the above-mentioned issues drives many researchers to develop tools that help facilitate the shucking of oysters.

Since ancient times, efforts have been made to create a more efficient method of shucking oysters, despite knowing that they are highly regarded for their taste. In fact, many different oyster shucking technologies have been employed. Nonetheless, there have been a significant growth in inventions and innovations in this subject during the past 50 years (Martin and Hill, 2006). An oyster clamp is one of the best example of these technologies that is usually a wooden or metal shucking tool design to open bivalves shellfish specifically, an oyster. The unique shape of this tool helps to easily locate and hold the oyster. The material of this tool is strong and durable that helps the user to open oysters without hurting your hands. It is considered a good tool for families, restaurants, shopping malls and supermarkets (Amazon Inc., 2024).

An invention related to a portable manual oyster opener was developed by Chi-Minh Le in 2009. It comprises of press, plunger, bar, splitter, gear and base. A known disadvantage of this tool is that it is limited to a single quantity production and lack of concern for the safety of the user thus, shucking process is done on only one oyster per process and the splitter is very expose from the hand of the user. Thereby, requiring more time for the procedure and caution when using the tool.

Gregory Adams Rotello, (2011) fabricated a hand-held oyster-opening device herein described simple, quick and ultra-safe to operate. However, the device has low efficiency caused by the single used operation. In addition, the materials of the device are seem to be expensive and might not be suitable for the budget of all buyers and operators.

Enhancing the equipment' capacity for processing tasks which goes beyond only to small quantity production and takes user efficiency and safety into account is essential for recovering the oyster shucking process. Therefore, the purpose of this project is to develop, construct, and assess the performance of the dual oyster shucking clamp.

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(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned. .)

A developmental type of research will be used in this study, wherein an existing oyster clamp device for oyster shucking use will be re-designed, fabricated, and evaluated. The development will include the addition of a dual set of shucking operations, the addition of a safety handle and hinges, and optimizing the speed of production.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The general objective of this study was to modify and develop a dual oyster shucking clamp device for oyster opening use. Specifically, this study aims to:

1. Modify and fabricate the existing oyster shucking clamp for dual oyster opening operation;
2. Optimize the device in terms of the processing speed, and the safety of the user in carrying out the process; and
3. Perform technology valuation using R&D cost.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

US7785176B1 Portable manual oyster opener and method

The present invention is an apparatus which manually forces open shells during the downward stroke of a reciprocating plunger. Opening can occur through shearing of the two shell halves along their seam. The tip of the plunger can be a long inclined blade and wedge shaped. The apparatus may open all sizes of shells by applying a downward force on at least one of the shell halves. Alternatively, both shearing and sliding forces are applied. In one embodiment the shell is placed parallel to the angle of taper of the tip for opening. When the plunger comes down it exerts by frictional shearing force on the shell which causes the halves to slide off each other and open providing access to the inside.

US7455576B1 Oyster splitting device and method of use

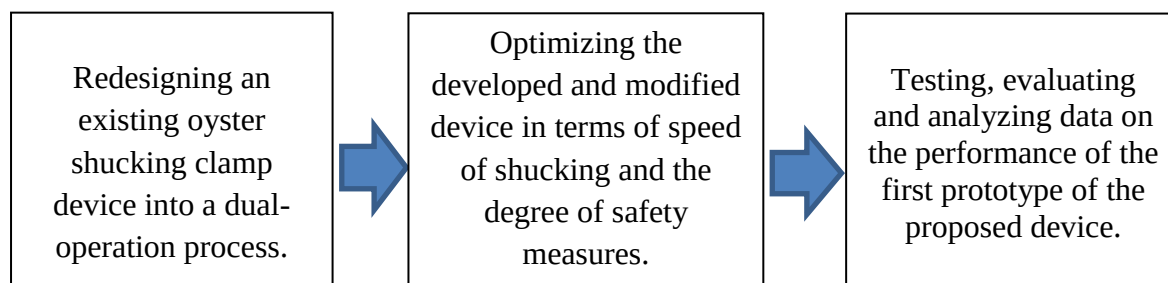
The device refers to an oyster splitting devices split open oysters and clams. A bar clamp has a drawbar slidably inserted through a grip. The drawbar has a stop on one end and a jaw on its other end. The grip has a jaw trigger and a bar release trigger pivotably attached to it. A spike is attached to the grip with the spike's pointed end facing the jaw. An oyster's open end is placed against the jaw. When the jaw trigger is pulled by the user, the jaw urges the hinged end of the oyster against the spike, splitting its shells apart. The drawbar's opposing end opposite the jaw may be tempered and tapered and used to pry oysters from rocks and/or remove meat from split oysters. A safety cover connected to the grip by a lanyard removably covers the opposing end of the drawbar opposite the jaw when it is not in use.

20230180773 Oyster Shucking Apparatus

The invention relates to an oyster clamp comprising a base element having an indentation for receiving an oyster to be opened. A lip coupled to the base element hangs over and impinges upon an edge of a work surface on which the oyster clamp is placed, and a slide inhibitor is arranged to prevent the base element from sliding on the work surface. A backstop is affixed to the base element to prevent the oyster from exiting the back of the oyster clamp when in use. An oyster clamping element is operatively coupled to a handle and to the base element, such that operating the handle causes the clamping element to push the oyster down against the base element.

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(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.)



13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Research and Sampling Design

The study is a developmental type of research. It shall include three stages such as design modification, optimization, and performance evaluation.

Design Modifications

The existing device relates to a single-operated oyster shucking clamp (Figure 1.a). The device may be modified by:

1. adding a safety handle to improve the device's ability to protect users from harm and address operation-related problems;
2. adding another oyster holder for dual oyster shucking in a single operation process; and
3. adding hinges to the upper and lower base to allow adjustments in gripping of various sizes of oysters.

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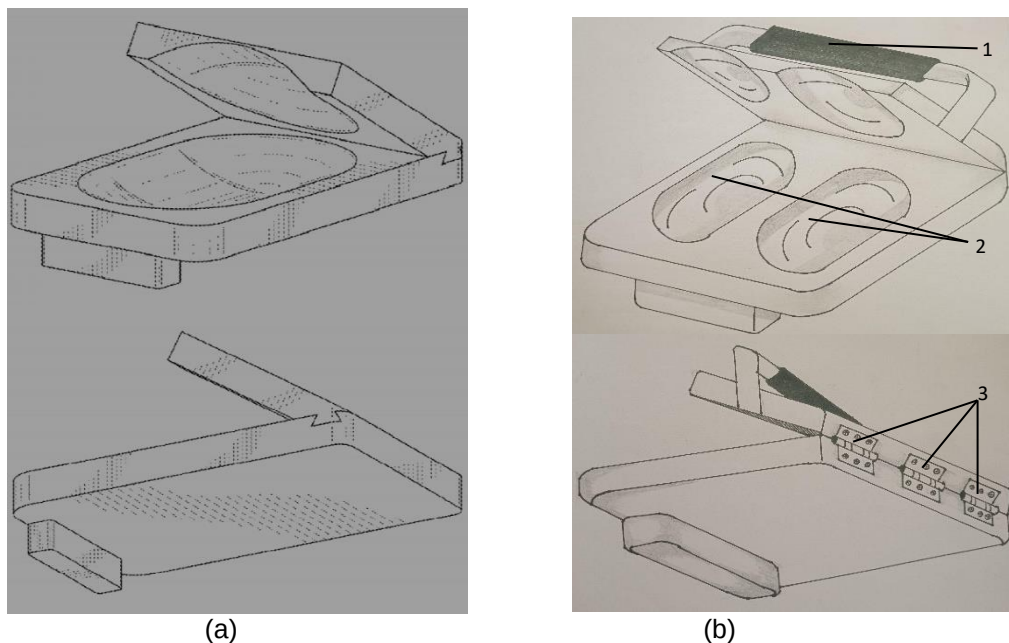


Figure 1. Perspective views of the existing oyster clamp (a) and the proposed device design (b)

Optimization

Device optimization will be conducted to determine the optimum values of the shucking speed and safety measures using the Design of Experiments (DOE), specifically the Response Surface Modelling (RSM).

The experiment will use the Central Composite Design (CCD), with three factors and three levels for each factor. The arrangements in the previous device will be used as the initial control treatment.

The experiment will be conducted in sixteen (4) runs for fifteen (15) minutes per run for the commodity. Data will be analyzed using the Analysis of Variance to be interpreted at 5% significance levels.

The optimum values in terms of efficiency will be used for final device design and fabrication.

Performance Evaluation

In the performance evaluation of the device, the following materials will be used: fresh oysters, shucker knife, caliper, and digital weighing scale, and a timer.

The following procedure will be followed in oyster shucking:

Selection. Newly harvested oysters of different sizes will be selected and used as test materials in this study. The fresh oysters will be divided into single shell samples and sorted based on size. Selected samples will be measured in length and width as well as in weight. Every measurement such as the weight and size of the test materials and the amount of time needed to complete the

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shucking procedure will be recorded in this study.

Separating. The test materials will be separated manually using a stainless knife and sorted.

Washing. The separated materials will be washed thoroughly with clean water.

Weighing. The washed materials will be weighed using digital weighing scale and caliper.

Shucking. The measured materials will be placed into the dual oyster clamp for shucking.

Technology Valuation

Technology cost will be computed in terms of R&D cost, considering expenses on personnel services and MOOE.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	Month 1	Month 2	Month 3	Month 4	Month5
1. Design Modification					
2. Optimization					
3. Performance Evaluation					
4. Technology Valuation					
5. Terminal report writing and finalization					

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

- 1 Patent Application
- 1 Product – Dual Oyster Shucking Clamp
- 1 IEC Material
- 1 People Services - Extension

(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

The study will develop a dual oyster shucking clamp device, which may be licensed for distribution or sale to an approved fabrication business organization. Small-medium scale processors who will adopt/buy the machine will be assisted to enroll to CAPSU ATBI or ABI. Similarly, trainings from extension activities will be conducted in different aspects.

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(17) Target Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

1. Oyster Farmers in the province of Capiz
2. Small-medium scale oyster shucking operators
3. Normal citizens who enjoys to eat oysters

(18) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization
Stephanie S. Pimentel, PhD.	Study Leader	Biology
Cedie A. Borreros	Member	Fisheries

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours): 3 hours per week

Program/Project Members (in hours): 3 hours per week

(19) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		
Fabricator	1X3 months	Php 54,000.00
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Oyster Supplies	5 sacks X 1 commodity X Php2,100/sack	Php10,500.00
- Construction Materials	10,000.00	Php10,000.00
- Miscellaneous (for incidental expenses)	5,000	Php5,000
C. CAPITAL OUTLAY		
Total		Php 79, 500.00

(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

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(21) Capsule Curriculum Vitae (One-page brief C.V. only)

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